

Web Application Programming and Hacking

Front-end Web Development Lab

Instructor: Dr. Phu Phung

Student: Connor Monaghan

Show my email



A simple HTML page

Using the [W3Schools template](#)

Lab1

This lab shows how to use Wireshark to analyze incoming and leaving messages on a network, specifically HTTP messages. This lab also teaches how to use Telnet for the same purpose. Next, we developed a program in C to print statements to a web client. After, we used PHP to write a hello world program and an echo program. We then analyzed the echo program and learned the difference between GET requests and POST requests.

Lab2

In this lab, we experimented with different HTML tags such as `<div>`, `<a>`, and `<br>`. We also learned how to insert images using `<img>` and we learned how to use forms with the `<form>` tag. We then learned how to use inline code using the `<script>` tag and we also learned how to write code in another .js file and include it in the HTML code to run. We were taught that using code from an outside source/link is not a good practice. Next, we explored how Ajax and jQuery can work together to make post and get

requests use less lines of code and be easier to write. We also used fetch to return a promise.

Hackathon

In this hackathon, I hacked 6 levels of different websites to input a code snippet to show an alert. The websites used techniques such as filtering to remove the `<script>` tag and different techniques of inputting the data such as GET or POST. For levels 2-6, I have guessed what the source code may look like in order to make hacking it more difficult. Then, I modified my echo.php program and my waph-monaghan.html file to include input validation and encoding. Screenshots showing the GitHub merges are at the bottom. This lab taught me how easy it can be for hackers to hack a website if it isn't secured properly. Now, I know how to secure my sites a little bit better from injection type attacks

Individual Project

This individual project is a public portfolio designed to showcase my skills and experience to potential employers. In addition to serving as my resume, this repository meets course requirements by including interactive scripts we learned in previous labs.